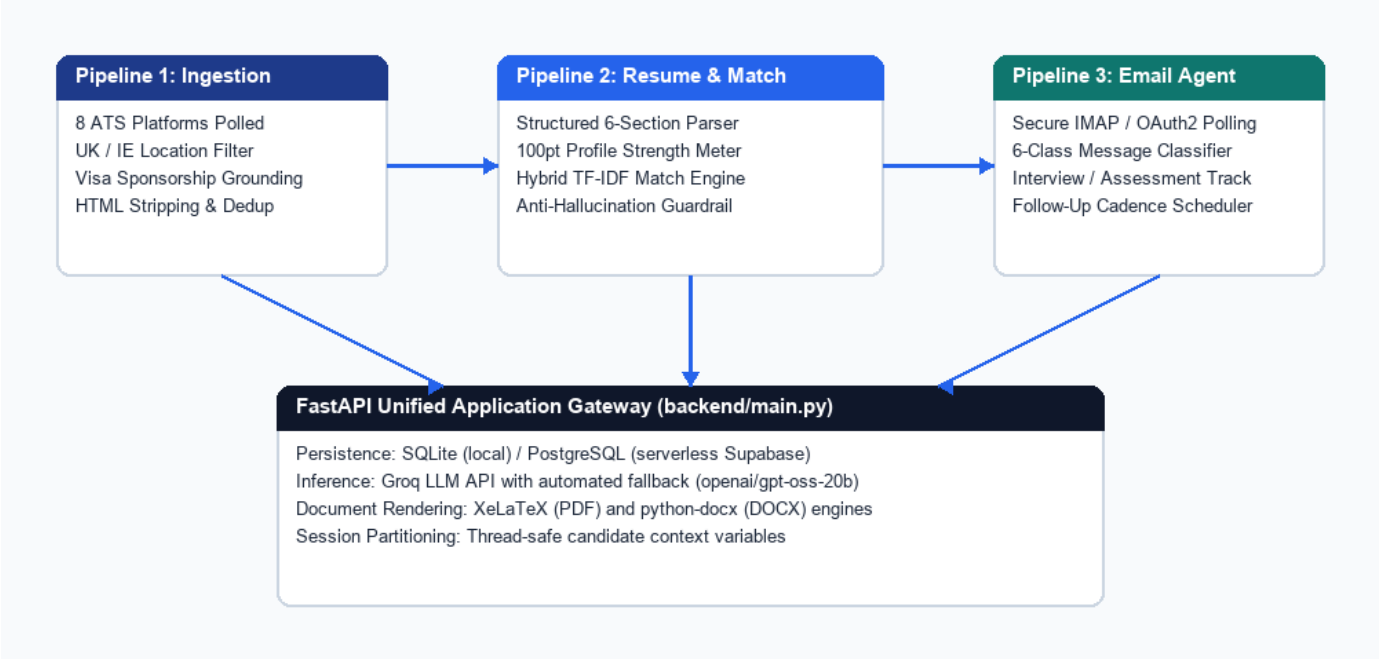
Cafy Agent Backend Architecture

End-to-End Technical Specifications, Flow Diagrams & Decision Trees

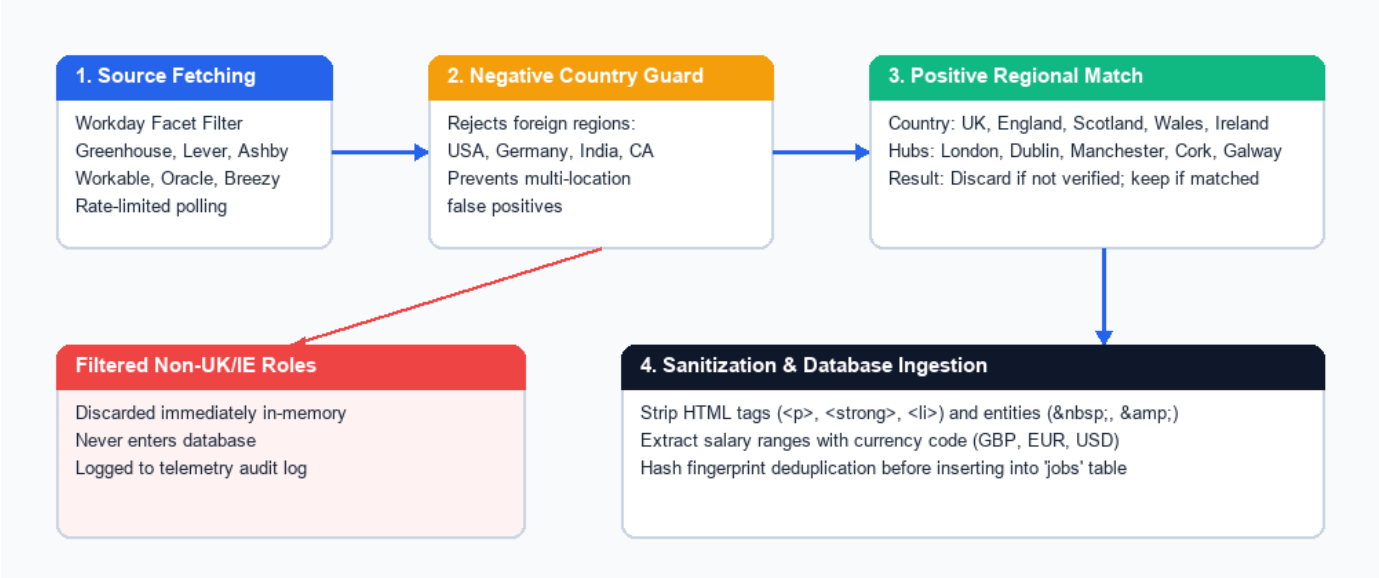
1. System Architecture Overview

Cafy Agent is structured around three core automation pipelines operating on an asynchronous FastAPI runtime. Pipeline 1 discovers, filters, and standardizes ATS jobs across the UK and Ireland. Pipeline 2 manages structured resume parsing, 100-point profile strength evaluation, hybrid job recommendations, and AI-tailored resume generation. Pipeline 3 provides autonomous email agent tracking, recruiter message classification, and follow-up scheduling.



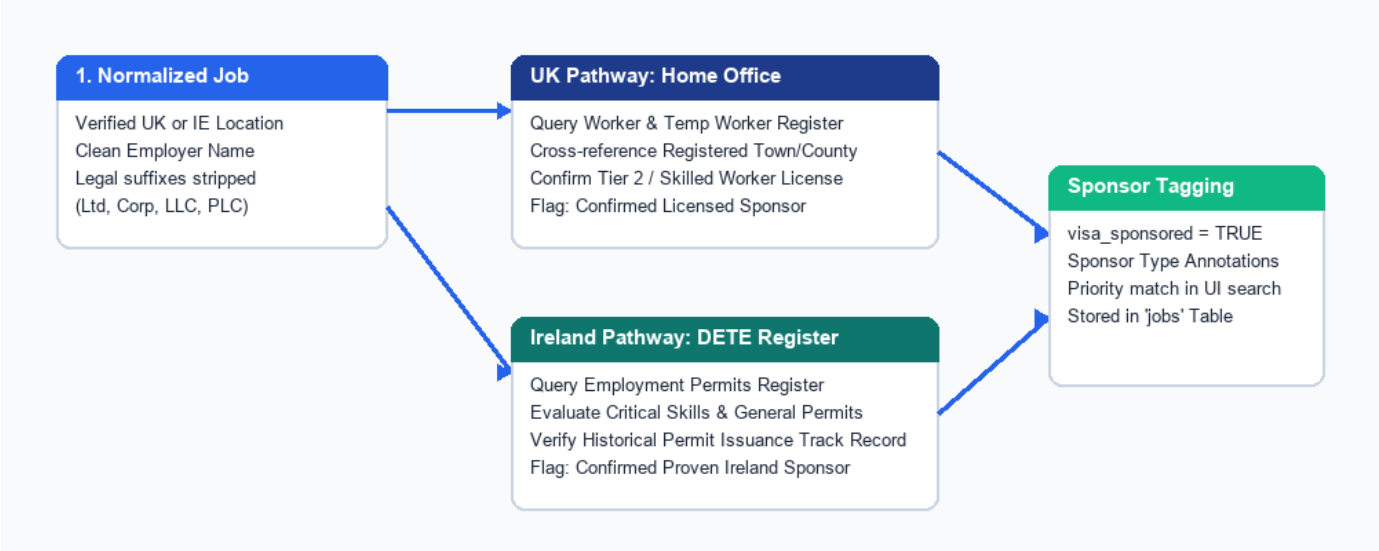
2. ATS Ingestion & Geographic Validation Logic

All incoming listings are strictly validated to enforce exclusive containment in the United Kingdom and Republic of Ireland:
1. Workday Facet Ingestion: Inquires tenant country facets over the wire to request only UK/IE jobs, discarding 85%+ of overseas roles.
2. Negative Country Blocklist: Immediate rejection if foreign countries (USA, Canada, Germany, India) appear in the location string.
3. Positive Regional Hubs: Validates regional markers (UK, Scotland, Wales, Ireland) and 35+ major employment cities (London, Dublin, Cork).
4. Sanitization & Dedup: Cleans raw HTML tags, formats compensation into standardized ranges, and eliminates duplicates via SHA-256.



3. Visa Sponsorship Verification Logic

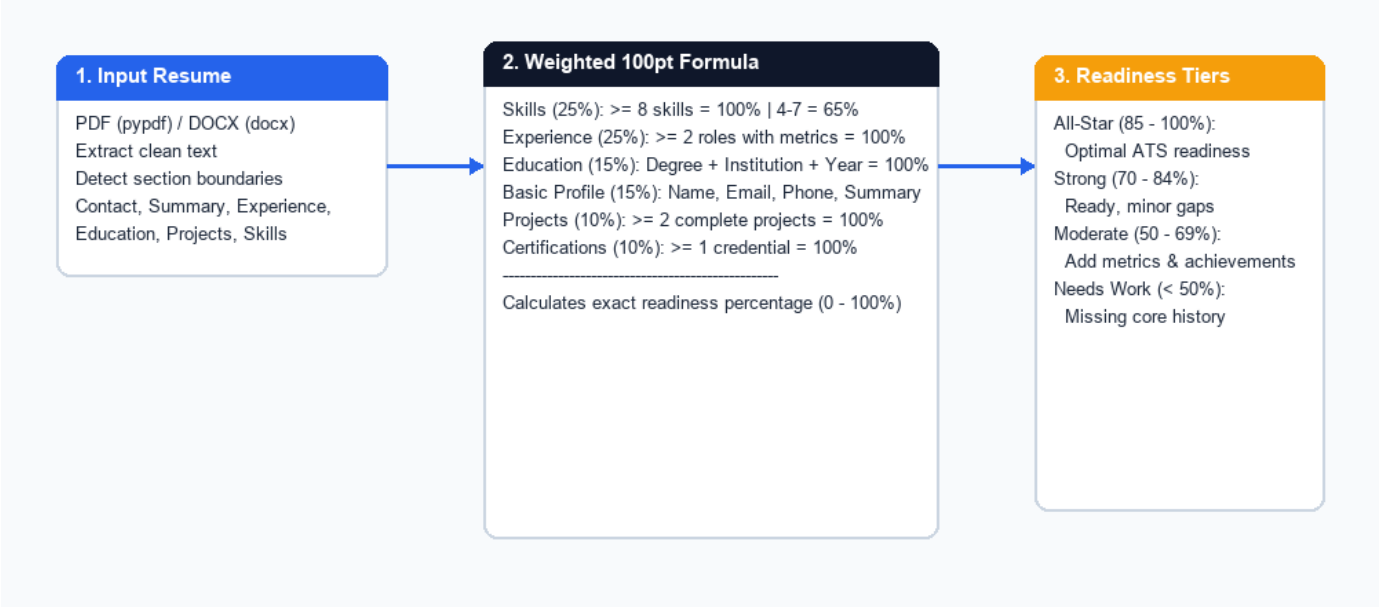
Jobs are cross-referenced against official government registries to verify work permit eligibility:
- UK Home Office Register: Strips corporate suffixes (Ltd, PLC, Corp) and matches against licensed Tier 2 / Skilled Worker employers.
- Irish DETE Register: Evaluates track records of issued Critical Skills and General Employment permits across Ireland.



4. Resume Parsing & Profile Strength Scoring

Resumes uploaded as PDF or DOCX are parsed into structured sections (Contact, Summary, Experience, Education, Projects, Skills). The profile is evaluated against an industry-calibrated 100-point completeness scale:

- Skills (25%): >= 8 validated technical skills awards full section points; 4-7 yields 65%.
- Experience (25%): >= 2 complete roles with dates and outcome-driven bullet points awards full points.
- Education (15%): Degree title, academic institution, and graduation year verified.
- Basic Details (15%): Candidate full name, valid email address, phone number, and professional summary.
- Projects & Certifications (20%): Evaluates hands-on project architectures and professional credentials.

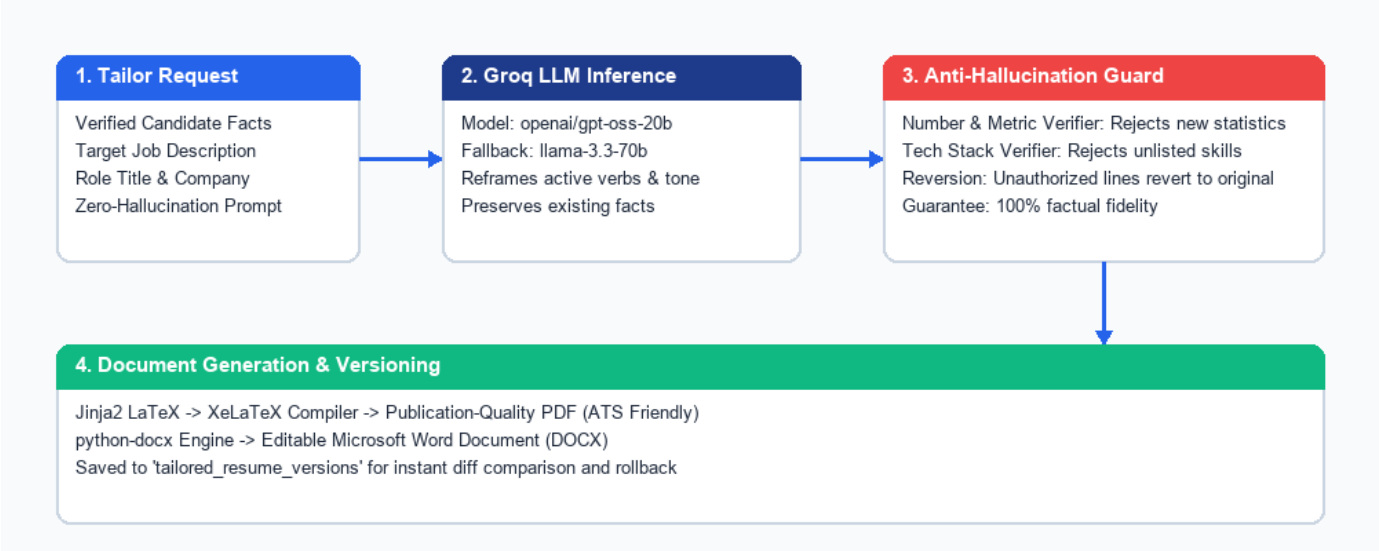


5. Hybrid Job Recommendation Algorithm

Candidate profiles are compared against stored jobs using a multi-factor score: 50% Skill Overlap (Jaccard and keyword intersection), 30% Seniority & Role Alignment (Junior, Mid, Senior, Lead semantics), and 20% Geographic Feasibility & Sponsorship requirements. Roles scoring >= 60% are recommended; scores >= 80% are highlighted as high-confidence matches.

6. AI Tailoring & Anti-Hallucination Guardrails

The resume tailoring engine optimizes wording for the target role while guaranteeing 100% factual truth:
- Inference Engine: Calls Groq LLM API with primary model openai/gpt-oss-20b and automatic fallback (llama-3.3-70b-versatile).
- Factual Invariance Guardrail: The model is prohibited from inventing employers, dates, achievements, or unlisted technologies.
- Metric Verifier: Any rewritten line introducing new numbers or metrics not present in the original resume is rejected.
- Dual Compilers: Outputs both publication-quality XeLaTeX PDF resumes and fully formatted Microsoft Word DOCX files.



7. Email Agent & Follow-Up Lifecycle

Pipeline 3 connects to candidate inboxes via IMAP/OAuth2 to automate recruiter communication and application status:
1. Classification: Analyzes recruiter emails into Interview Invitations, Technical Assessments, Acknowledgments, Offers, or Rejections.
2. State Advancement: Receiving an interview invite automatically moves the application from 'Applied' to 'Interviewing'.
3. Assessment Deadlines: Parses take-home test due dates and triggers high-priority reminders.
4. Cadence Automation: Detects applications with no reply after 7-10 business days and pre-drafts professional follow-up inquiries.

